GTU CSE 495 ve 496

P-wave and S-wave Detection Using Machine Learning

CSE 496
First Presentation

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- Project Design Plan
- Project Requirements
- Success Criteria
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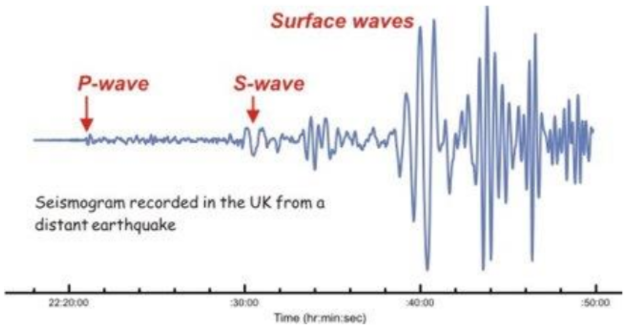
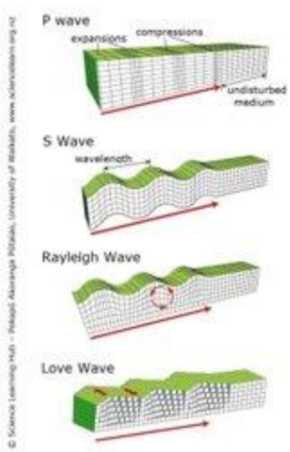
Scheme and Description of the Project





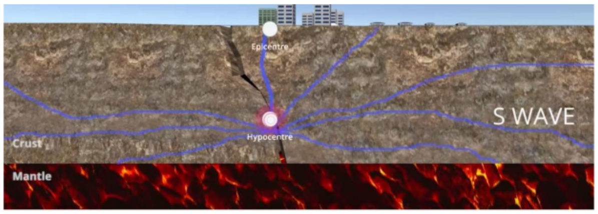
- When an earthquake occurs, the shockwaves of released energy that shake the Earth are called seismic waves.
- In earthquake waveforms, P-wave, S-wave and surface wave are the three main groups of wave trains. P-wave is the first wave group to arrive in an earthquake due to its faster propagation speed than S-wave and surface wave.

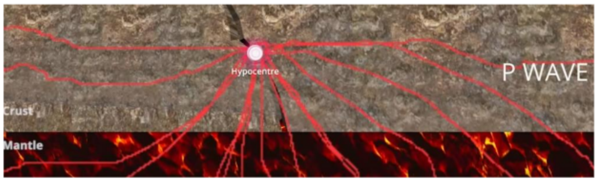
- So the projects aims to implement a machine learning model for p-wave and s-wave detection for the early detection of earthquakes to warn people.

Project Design Plan

- Define our dataset with seismic waves.
- Define our regression problem.
- Create ML models using picked algorithms.
- Test your ML model with test data.

- Compare the models.
- Improve the ML model





Project Requirements

- It needs a clean dataset for the model creation.
- If we can not find a clean dataset we can create a dataset using distributions.
- Python Machine Learning Libraries
- NumPy
- SciPy
- Scikit-learn
- TensorFlow
- Keras
- Regression Algorithms for predict the eartquake before it happeend.

Timeline

05.04.2023

- Searching about p-wave
- Searching dataset
- Look for similar projects

First Representation

17.05.2023

- Learning machine learn algorithms
- Creating a model and test

Mid Representation

21.06.2023

- Test
- Result
- Improving

Final Presentation

Success Criteria

- 65% accuracy detecting the P-wave.
- 75% accuracy detecting the S-wave.
- Use at least 5 years data. The data would be use STEAD or provided by the civil engineering dept:
- https://www.kaggle.com/datasets/omercolakoglu/turkey-earthquake-data-for-20-years
- https://github.com/smousavi05/STEAD

Resources

1. http://earthquakes.bgs.ac.uk/education/eq_guide/eq_booklet_how_we_measure.htm
2. https://ieeexplore.ieee.org/document/8871127
3. https://www.youtube.com/watch?v=RqqqSnaTfQo
4. https://www.sciencelearn.org.nz/resources/340-seismic-waves
5. https://www.sciencedirect.com/science/article/pii/S2405844021027080
6. https://www.frontiersin.org/articles/10.3389/feart.2022.1032839/full
7. https://se.copernicus.org/articles/10/1989/2019/
8. https://www.researchgate.net/publication/324744564_P_Wave_Arrival_Picking_and_First-Motion_Polarity_Determination_With_Deep_Learning